

## Subquery Assignment (50 questions)

#1 Find customers who made purchases above average sales amount

```
select c.customer_name from customers as c join sales as s using(customer_id) group by s.customer_id
having sum(s.total_amount)>
(select avg(total_spent) from
(select sum(ss.total_amount) as total_spent from sales as ss group by ss.customer_id) as temp);
```

#2 Find products priced above average price

```
select product_name from products where price>
(select avg(pp.price) from products as pp);
```

#3 Find employees earning more than average salary

```
select employee_name from employees where salary>
(select avg(salary) from employees);
```

#4 Find customers who made at least one purchase

```
select c.customer_name from customers c where c.customer_id in
(select s.customer_id from sales s);
```

#5 Find products that were never sold

```
select p.product_name from products p where p.product_id not in
(select si.product_id from sale_items si);
```

#6 Find customers who never made a purchase

```
select c.customer_name from customers c where c.customer_id not in
(select s.customer_id from sales s);
```

#7 Find the highest priced product

```
select product_name from products where price=
(select max(p2.price) from products p2);
```

#8 Find the lowest priced product

```
select product_name from products where price=
(select min(p2.price) from products p2);
```

#9 Find employees with the highest salary

```
select employee_name from employees where salary =
(select max(e.salary) from employees e);
```

#10 Find sales with amount greater than average

```
select sale_id from sales where total_amount>
(select avg(s.total_amount) from sales s);
```

#11 Find products priced below average

```
select product_name from products where price<
(select avg(pp.price) from products as pp);
```

#12 Find customer who spent the most

```
select c.customer_name from customers c join sales s using (customer_id) group by c.customer_name
having sum(s.total_amount)=
(select max(total_spent) from (select sum(s2.total_amount) as total_spent from sales s2 group by
s2.customer_id)as temp);
```

#13 Find products sold more than average quantity

```
select p.product_name from products p join sale_items si using(product_id) group by
p.product_id,p.product_name having sum(si.quantity)>
(select avg(total_quantity) from (select sum(si2.quantity) as total_quantity from sale_items si2 group by
si2.product_id)as temp);
```

#14 Find employees who handled above average sales

```
select e.employee_name from employees e join sales s using(employee_id) group by
e.employee_id,e.employee_name having sum(s.total_amount)>
(select avg(total_spent) from (select sum(s2.total_amount) as total_spent from sales s2 group by
s2.employee_id)as temp);
```

#15 Find customers who spent more than average

```
select c.customer_name from customers c join sales s using(customer_id) group by
c.customer_id,c.customer_name having sum(s.total_amount)>
(select avg(total_spent) from (select sum(s2.total_amount) as total_spent from sales s2 group by
s2.customer_id)as temp);
```

#16 Find products with stock greater than average

```
select product_name from products where stock_quantity>
(select avg(p.stock_quantity) from products p);
```

#17 Find suppliers who supplied expensive products

```
select supplier_name from suppliers join products using(supplier_id) where price=
(select max(p.price) from products p);
```

#18 Find categories that contain expensive products

```
select category_name from categories join products using(category_id) where price=
(select max(p.price) from products p);
```

#19 Find customers who bought expensive products

```
select c.customer_name from customers c join sales s using(customer_id) join sale_items si
using(sale_id) join products p using(product_id)
where p.price=(select max(p2.price) from products p2);
```

#20 Find employees who handled high value sales

```
select e.employee_name from employees e join sales s using(employee_id) group by
e.employee_id,e.employee_name having sum(s.total_amount)=
(select max(amount) from (select sum(ss.total_amount) as amount from sales ss group by
ss.employee_id) as temp);
```

#21 Find products with maximum stock

select product\_name from products where stock\_quantity =  
(select max(p.stock\_quantity) from products p);

#22 Find products with minimum stock  
select product\_name from products where stock\_quantity =  
(select min(p.stock\_quantity) from products p);

#23 Find customers with earliest join date  
select customer\_name from customers where join\_date =  
(select min(c.join\_date) from customers c);

#24 Find customers with latest join date  
select customer\_name from customers where join\_date =  
(select max(c.join\_date) from customers c);

#25 Find sales with highest amount  
select \* from sales where total\_amount =  
(select max(s.total\_amount) from sales s);

#26 Find sales with lowest amount  
select \* from sales where total\_amount =  
(select min(s.total\_amount) from sales s);

#27 Find products that were returned  
select product\_name from products where product\_id in  
(select product\_id from returns);

#28 Find products that were never returned  
select product\_name from products where product\_id not in  
(select product\_id from returns);

#29 Find customers who returned products  
select customer\_name from customers join sales using(customer\_id) where sale\_id in  
(select sale\_id from returns);

#30 Find employees involved in returns  
select employee\_name from employees join sales using(employee\_id) where sale\_id in  
(select sale\_id from returns);

#31 Find categories that have sold products  
select distinct category\_name from categories join products using(category\_id) where product\_id in  
(select product\_id from sale\_items);

#32 Find suppliers who supplied returned products  
select distinct supplier\_name from suppliers join products using(supplier\_id) where product\_id in  
(select product\_id from returns);

#33 Find products with total quantity sold above average

```
select product_name from products join sale_items using(product_id)
group by product_id,product_name having sum(quantity)>
(select avg(total_quantity) from (select avg(quantity) as total_quantity from sale_items group by
product_id) as temp);
```

#34 Find employees earning below average salary

```
select employee_name from employees where salary<
(select avg(e.salary) from employees e);
```

#35 Find customers with membership above average level

```
select customer_name from customers where membership_id>
(select avg(c.membership_id) from customers c);
```

#36 Find suppliers from countries having more than one supplier

```
select supplier_name from suppliers where country in
(select country from suppliers group by country having count(*)>1);
```

#37 Find customers from countries having more than two customers

```
select customer_name from customers where country in
(select country from customers group by country having count(*)>2);
```

#38 Find products belonging to popular categories

```
select product_name from products where category_id in
(select category_id from products group by category_id having count(*)>2);
```

#39 Find employees working in roles with more than one employee

```
select employee_name from employees where role in
(select role from employees group by role having count(*)>1);
```

#40 Find sales that contain more than one item

```
select* from sale_items where sale_id in
(select sale_id from sale_items group by sale_id having count(*)>1);
```

#41 Find products with highest quantity sold

```
select product_name from products join sale_items using(product_id) group by
product_id,product_name having sum(quantity)=
(select max(total_quantity) from (select sum(si.quantity) as total_quantity from sale_items si group by
si.product_id)as temp);
```

#42 Find customers with highest number of orders

```
select c.customer_name from customers c join sales s using(customer_id) group by
c.customer_id,c.customer_name having count(s.customer_id) =
(select max(max_customer_id) from (select count(ss.customer_id) as max_customer_id from sales ss
group by ss.customer_id)as temp);
```

#43 Find employees with most sales handled

```
select employee_name from employees e join sales s using(employee_id) group by
e.employee_id,e.employee_name having count(s.employee_id)in
```

(select max(total\_order) from (select count(s.employee\_id) as total\_order from sales s group by s.employee\_id)as temp);

#44 Find products whose price is greater than all other products

select product\_name from products where price =  
(select max(p.price) from products p);

#45 Find products whose price is lower than all other products

select product\_name from products where price =  
(select min(p.price) from products p);

#46 Find customers who never returned any product

select customer\_name from customers where customer\_id not in  
(select customer\_id from sales join returns using(sale\_id));

#47 Find suppliers who have no products

select supplier\_name from suppliers where supplier\_id not in  
(select supplier\_id from products);

#48 Find products that were never restocked

select product\_name from products where product\_id not in  
(select product\_id from inventory);

#49 Find sales where amount is above average for that customer

select\*from sales s where s.total\_amount>  
(select avg(s2.total\_amount) from sales s2 where s.customer\_id=s2.customer\_id);

#50 Find employees earning in top 3 salaries

select employee\_name from employees where salary in  
(select salary from (select distinct salary from employees order by salary desc limit 3)as tempt);